

ELIAS HAKENSO

AI / ML : Model Evaluation, Fine-Tuning, RAG, LLM Applications

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SUMMARY

AI/ML practitioner focused on model output evaluation and LLM application development. Built and evaluated LLM systems end to end, including label taxonomy design, data annotation, fine-tuning and benchmarking classifiers, and analysis of failure modes and confidence calibration. Works independently against detailed specs and produces clear, structured written analysis. 3+ years of professional data work, plus a BS in CIT and an in-progress MS in Data Science.

AI / ML PROJECTS

Provenance Guard: AI Content Detection & Confidence Scoring *Flask · Groq (Llama-3.3-70b) · Python stylometry*

- Built a backend that classifies whether text is human or AI written using two independent detectors: a Groq LLM semantic classifier, and a pure Python stylometric heuristic (burstiness, type-token ratio, punctuation density), fused into a single calibrated confidence score across three categories.
- Designed the scoring logic around evaluation tradeoffs: a deliberately wide “uncertain” band and a disagreement penalty that pulls conflicting signals toward 0.5, explicitly minimizing the costliest error of falsely flagging human work as AI.
- Documented a reproducible false positive failure mode (formal writing and writing by non-native English speakers misread as AI) and shipped an appeals workflow that routes contested calls to human review. Logged every decision as structured JSON for auditability, with per-IP rate limiting to protect integrity and API cost.

TakeMeter: Fine-Tuned Text Classifier & Model Evaluation *Python · Hugging Face · DistilBERT · Groq*

- Annotated a balanced 200 example dataset under a mutually exclusive three class taxonomy, writing explicit decision rules for borderline cases before labeling to keep annotation consistent.
- Fine-tuned distilbert-base-uncased and benchmarked it against a zero-shot llama-3.3-70b baseline using precision, recall, F1, and confusion matrix analysis on a held out test set.
- Diagnosed why the fine-tuned model lost to the baseline (a small model on roughly 140 examples learned surface form rather than reasoning), tracing systematic misclassifications to a labeling boundary limitation rather than annotation noise, and specifying what data would fix it.

The Unofficial Guide: Retrieval-Augmented (RAG) Q&A System *Python · ChromaDB · sentence-transformers · Groq · Gradio*

- Built a RAG question answering system over off-campus student housing documents, using a two-tier chunking strategy and cosine distance retrieval from a ChromaDB vector store to ground answers in source text.
- Evaluated the system on a fixed question set to assess retrieval relevance and answer quality, and reduced hallucination by constraining responses to retrieved context. Deployed through a Gradio interface.

FitFindr: Multi-Tool AI Agent *Python · Groq · Gradio · pytest*

- Built an agentic application with a conditional planning loop that routes requests through chained tools (search, outfit suggestion, caption generation), branching on tool output rather than a fixed sequence, with shared session state and per-tool error handling for empty results and malformed inputs. Covered tool behavior with pytest.

EXPERIENCE

Volunteer Web Developer

Dec 2025 to Present

Hasiet's Vision International, Seattle, WA

- Support a nonprofit around 20 hrs/week with website redesign, maintenance, and content updates, collaborating with leadership to keep content accurate and accessible.

Data Entry Specialist

Dec 2021 to May 2025

Minnesota State University, Mankato, MN

- Automated data cleaning and reporting with Python and Excel across 25,000+ student records, and applied statistical analysis in SAS and Excel to generate actionable insights for university departments.

EDUCATION

M.S. in Data Science (Executive), University of the Cumberland

Expected May 2027

CodePath AI201 (in progress)

Expected Aug 2026

- Completed four take-home AI projects spanning RAG, LLM agents, model fine-tuning, and model output evaluation.

B.S. in Computer Information Technology, Minor in Mathematics, MSU Mankato

Aug 2025

GPA 3.7 · Magna Cum Laude

SKILLS

AI / ML Evaluation: model output evaluation, data annotation and label taxonomy design, precision/recall/F1, confusion matrix analysis, baseline benchmarking, confidence calibration

LLM & AI Engineering: LLM APIs (Groq), prompt design, agentic workflows, RAG (ChromaDB, sentence-transformers), fine-tuning (Hugging Face, DistilBERT), stylometry

Data & Analytics: Python (pandas, NumPy, scikit-learn), SQL, SAS, Excel, statistical analysis, dashboards, stakeholder communication

Languages & Tools: Python, SQL, JavaScript, Java, C#, Kotlin, Dart; Git, Flask, Gradio, pytest, Hugging Face, React, Node.js, Express, MongoDB